

Chapter 5: Inner Product Spaces

Chapter overview and learning map

Inner products turn vector spaces into geometric objects. In this chapter, the ideas develop in the following order:

length and dot product $\longrightarrow$ inner products $\longrightarrow$ orthonormal bases $\longrightarrow$ projection and least squares $\longrightarrow$ cross product

The examples below use original numerical values and selected themes such as the Spurs, hiking, badminton, driving, travel, and music.

1 Length and Dot Product in $\mathbb{R}^n$

Norm and unit vector

For $\mathbf{v} = (v_1, \dots, v_n)$,

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}.$$

A vector is a unit vector when its norm is 1. For $\mathbf{v} \neq \mathbf{0}$, the unit vector in its direction is $\hat{\mathbf{v}} = \mathbf{v} / \|\mathbf{v}\|$.

Spurs-inspired norm calculation

Let $\mathbf{v} = (9, 12, 20)$. Then

$$\begin{aligned}\|\mathbf{v}\| &= \sqrt{9^2 + 12^2 + 20^2} \\ &= \sqrt{81 + 144 + 400} \\ &= \sqrt{625} = 25.\end{aligned}$$

Therefore,

$$\hat{\mathbf{v}} = \frac{1}{25}(9, 12, 20) = \left(\frac{9}{25}, \frac{12}{25}, \frac{4}{5}\right).$$

A direct check gives

$$\|\hat{\mathbf{v}}\|^2 = \frac{81 + 144 + 400}{625} = 1.$$

Distance

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

For $\mathbf{u} = (3, -2, 5)$ and $\mathbf{v} = (9, 1, -3)$,

$$\mathbf{u} - \mathbf{v} = (-6, -3, 8), \quad d(\mathbf{u}, \mathbf{v}) = \sqrt{36 + 9 + 64} = \sqrt{109}.$$

Dot product, angle, and orthogonality

For $\mathbf{u} = (u_1, \dots, u_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$,

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + \dots + u_n v_n.$$

For nonzero vectors,

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

The vectors are orthogonal when $\mathbf{u} \cdot \mathbf{v} = 0$.

Let $\mathbf{a} = (4, 1, -2)$ and $\mathbf{b} = (2, -3, 5)$. Then

$$\mathbf{a} \cdot \mathbf{b} = 8 - 3 - 10 = -5, \quad \|\mathbf{a}\| = \sqrt{21}, \quad \|\mathbf{b}\| = \sqrt{38},$$

so

$$\cos \theta = -\frac{5}{\sqrt{798}}.$$

For the badminton movement vectors $\mathbf{p} = (3, 4, -1)$ and $\mathbf{q} = (4, -3, 0)$,

$$\mathbf{p} \cdot \mathbf{q} = 12 - 12 + 0 = 0,$$

so the movement directions are orthogonal in this teaching model.

Three important results

For all vectors $\mathbf{u}, \mathbf{v}$,

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\| \quad (\text{Cauchy-Schwarz}),$$

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \quad (\text{triangle inequality}).$$

Also, $\mathbf{u} \perp \mathbf{v}$ if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$

The dot product is also the matrix product

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

2 Inner Product Spaces

Inner-product axioms

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalar c , an inner product satisfies

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \langle \mathbf{v}, \mathbf{u} \rangle, \\ \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle, \\ \langle c\mathbf{u}, \mathbf{v} \rangle &= c \langle \mathbf{u}, \mathbf{v} \rangle, \\ \langle \mathbf{v}, \mathbf{v} \rangle &\geq 0,\end{aligned}$$

with $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ only when $\mathbf{v} = \mathbf{0}$.

Weighted inner product

Define on $\mathbb{R}^2$

$$\langle \mathbf{u}, \mathbf{v} \rangle_W = 2u_1v_1 + 5u_2v_2.$$

Symmetry and linearity follow from ordinary real-number operations. Positive definiteness follows from

$$\langle \mathbf{v}, \mathbf{v} \rangle_W = 2v_1^2 + 5v_2^2 \geq 0,$$

with equality only when $v_1 = v_2 = 0$. For $\mathbf{u} = (3, -1)$ and $\mathbf{v} = (2, 4)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle_W = 2(3)(2) + 5(-1)(4) = -8, \quad \|\mathbf{u}\|_W = \sqrt{18 + 5} = \sqrt{23}.$$

The expression $u_1v_1 - u_2v_2$ fails positive definiteness because

$$\langle (0, 1), (0, 1) \rangle = -1.$$

Polynomial and function inner products

For $p = a_0 + a_1x + a_2x^2$ and $q = b_0 + b_1x + b_2x^2$, define

$$\langle p, q \rangle = a_0b_0 + 2a_1b_1 + 3a_2b_2.$$

For $p = 1 + 2x - x^2$ and $q = 3 - x + 2x^2$,

$$\langle p, q \rangle = 3 - 4 - 6 = -7.$$

On $C[a, b]$, a standard function inner product is

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx.$$

For example, on $[0, 1]$,

$$\langle 1, x \rangle = \int_0^1 x dx = \frac{1}{2}.$$

Projection onto a vector

For $\mathbf{v} \neq \mathbf{0}$,

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}.$$

The residual $\mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$ is orthogonal to $\mathbf{v}$.

Detailed projection

Let $\mathbf{u} = (6, 3, 3)$ and $\mathbf{v} = (1, 2, 2)$. Then

$$\mathbf{u} \cdot \mathbf{v} = 6 + 6 + 6 = 18, \quad \mathbf{v} \cdot \mathbf{v} = 1 + 4 + 4 = 9.$$

Thus

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{18}{9}(1, 2, 2) = (2, 4, 4).$$

The residual is

$$\mathbf{r} = (6, 3, 3) - (2, 4, 4) = (4, -1, -1),$$

and

$$\mathbf{r} \cdot \mathbf{v} = 4 - 2 - 2 = 0.$$

3 Orthonormal Bases and Gram–Schmidt

Orthogonal and orthonormal sets

A set is orthogonal when distinct vectors have inner product zero. It is orthonormal when it is orthogonal and every vector has norm one. Every orthogonal set of nonzero vectors is linearly independent.

Consider

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0), \quad \mathbf{u}_3 = (0, 0, 1).$$

The vectors are pairwise orthogonal and all have norm one, so they form an orthonormal basis of $\mathbb{R}^3$.

For $\mathbf{w} = (9, 12, 20)$, the coordinates relative to this basis are

$$\begin{aligned} c_1 &= \mathbf{w} \cdot \mathbf{u}_1 = \frac{21}{\sqrt{2}}, \\ c_2 &= \mathbf{w} \cdot \mathbf{u}_2 = -\frac{3}{\sqrt{2}}, \\ c_3 &= \mathbf{w} \cdot \mathbf{u}_3 = 20. \end{aligned}$$

Therefore,

$$[\mathbf{w}]_B = \begin{bmatrix} 21/\sqrt{2} \\ -3/\sqrt{2} \\ 20 \end{bmatrix}.$$

Complete Gram–Schmidt calculation in $\mathbb{R}^3$

Start with

$$\mathbf{v}_1 = (1, 1, 0), \quad \mathbf{v}_2 = (1, 0, 1), \quad \mathbf{v}_3 = (0, 1, 1).$$

First,

$$\mathbf{w}_1 = \mathbf{v}_1 = (1, 1, 0).$$

Next,

$$\text{proj}_{\mathbf{w}_1} \mathbf{v}_2 = \frac{1}{2}(1, 1, 0),$$

so

$$\mathbf{v}_2 - \text{proj}_{\mathbf{w}_1} \mathbf{v}_2 = \left(\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

Scale this vector to use

$$\mathbf{w}_2 = (1, -1, 2).$$

For the third vector,

$$\text{proj}_{\mathbf{w}_1} \mathbf{v}_3 = \frac{1}{2}(1, 1, 0), \quad \text{proj}_{\mathbf{w}_2} \mathbf{v}_3 = \frac{1}{6}(1, -1, 2).$$

Hence

$$\begin{aligned} \mathbf{v}_3 - \text{proj}_{\mathbf{w}_1} \mathbf{v}_3 - \text{proj}_{\mathbf{w}_2} \mathbf{v}_3 &= (0, 1, 1) - \left(\frac{1}{2}, \frac{1}{2}, 0 \right) - \left(\frac{1}{6}, -\frac{1}{6}, \frac{1}{3} \right) \\ &= \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right). \end{aligned}$$

Use $\mathbf{w}_3 = (-1, 1, 1)$. Normalization gives

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1, 0), \quad \mathbf{u}_2 = \frac{1}{\sqrt{6}}(1, -1, 2), \quad \mathbf{u}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1).$$

Orthonormal basis for a solution space

The system

$$-x_1 + x_2 + x_3 = 0, \quad x_1 + x_4 = 0$$

has basis

$$\mathbf{v}_1 = (1, 1, 0, -1), \quad \mathbf{v}_2 = (1, 0, 1, -1).$$

Since $\mathbf{v}_1 \cdot \mathbf{v}_2 = 2$ and $\mathbf{v}_1 \cdot \mathbf{v}_1 = 3$,

$$\mathbf{w}_2 = \mathbf{v}_2 - \frac{2}{3}\mathbf{v}_1 = \frac{1}{3}(1, -2, 3, -1).$$

Thus an orthonormal basis is

$$\left\{ \frac{1}{\sqrt{3}}(1, 1, 0, -1), \frac{1}{\sqrt{15}}(1, -2, 3, -1) \right\}.$$

4 Mathematical Models and Least Squares Analysis

Least squares and normal equations

For $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, find $\hat{\mathbf{x}}$ minimizing

$$\|A\mathbf{x} - \mathbf{b}\|.$$

Every least-squares solution satisfies

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

The residual $\mathbf{r} = \mathbf{b} - A\hat{\mathbf{x}}$ satisfies $A^T \mathbf{r} = \mathbf{0}$.

Orthogonal complement

Let

$$S = \text{span}\{(1, 1, 0, 1), (1, 0, 1, 1)\}.$$

A vector belongs to $S^\perp$ when

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix} \mathbf{x} = \mathbf{0}.$$

Solving gives

$$S^\perp = \text{span}\{(-1, 1, 1, 0), (-1, 0, 0, 1)\}.$$

Projection onto a plane

The vectors $\mathbf{w}_1 = (1, 1, 0)$ and $\mathbf{w}_2 = (1, -1, 2)$ are orthogonal. For $\mathbf{v} = (9, 12, 20)$,

$$\begin{aligned} \text{proj}_S \mathbf{v} &= \frac{\mathbf{v} \cdot \mathbf{w}_1}{\mathbf{w}_1 \cdot \mathbf{w}_1} \mathbf{w}_1 + \frac{\mathbf{v} \cdot \mathbf{w}_2}{\mathbf{w}_2 \cdot \mathbf{w}_2} \mathbf{w}_2 \\ &= \frac{21}{2} \mathbf{w}_1 + \frac{37}{6} \mathbf{w}_2 \\ &= \left(\frac{50}{3}, \frac{13}{3}, \frac{37}{3} \right). \end{aligned}$$

The residual is

$$\mathbf{r} = \frac{23}{3}(-1, 1, 1),$$

and it is orthogonal to both $\mathbf{w}_1$ and $\mathbf{w}_2$.

Four fundamental subspaces

Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix}.$$

Row reduction gives

$$\text{rref}(A) = \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$\text{rank}(A) = 2,$$

$$\text{Row}(A) = \text{span}\{(1, 0, -2, -1), (0, 1, 1, 1)\},$$

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \right\},$$

$$N(A) = \text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\},$$

and

$$N(A^T) = \text{span} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Detailed least-squares regression line

Use the fictional driving data

$$(1, 2), (2, 3), (3, 5), (4, 4), (5, 7).$$

For $y = c_0 + c_1x$,

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \\ 4 \\ 7 \end{bmatrix}.$$

Then

$$A^T A = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 21 \\ 75 \end{bmatrix}.$$

The normal equations are

$$\begin{aligned} 5c_0 + 15c_1 &= 21, \\ 15c_0 + 55c_1 &= 75. \end{aligned}$$

Multiply the first equation by 3 and subtract:

$$10c_1 = 12, \quad c_1 = \frac{11}{10}.$$

Then

$$5c_0 + \frac{33}{2} = 21, \quad c_0 = \frac{9}{10}.$$

Therefore,

$$\hat{y} = \frac{9}{10} + \frac{11}{10}x.$$

The residual is

$$\mathbf{r} = \begin{bmatrix} 0 \\ -1/10 \\ 4/5 \\ -13/10 \\ 3/5 \end{bmatrix}, \quad A^T \mathbf{r} = \mathbf{0}.$$

For the fictional travel data $(0, 3), (1, 6), (2, 11), (3, 18), (4, 28)$, the quadratic least-squares model is

$$\hat{y} = \frac{108}{35} + \frac{57}{35}x + \frac{8}{7}x^2,$$

with residual sum of squares $4/35$.

5 Applications of Inner Product Spaces

Cross product

For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$,

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

It is orthogonal to both vectors. Its norm is the parallelogram area.

For $\mathbf{u} = (4, 1, 2)$ and $\mathbf{v} = (1, 3, -1)$,

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \mathbf{i}(1(-1) - 2(3)) - \mathbf{j}(4(-1) - 2(1)) + \mathbf{k}(4(3) - 1(1)) \\ &= (-7, 6, 11). \end{aligned}$$

The checks are

$$(-7, 6, 11) \cdot (4, 1, 2) = 0, \quad (-7, 6, 11) \cdot (1, 3, -1) = 0.$$

The parallelogram area is $\sqrt{206}$ and the triangle area is $\sqrt{206}/2$.

Continuous least-squares approximation

Approximate $f(x) = x^2$ on $[0, 1]$ by $g(x) = a + bx$. Orthogonality of the error to 1 and x gives

$$\int_0^1 (x^2 - a - bx) dx = 0,$$

$$\int_0^1 x(x^2 - a - bx) dx = 0.$$

Thus

$$\begin{aligned} a + \frac{1}{2}b &= \frac{1}{3}, \\ \frac{1}{2}a + \frac{1}{3}b &= \frac{1}{4}. \end{aligned}$$

Multiply the second equation by 2 and subtract the first:

$$\frac{1}{6}b = \frac{1}{6}, \quad b = 1.$$

Then $a = -1/6$. Therefore,

$$g(x) = x - \frac{1}{6}.$$

Fourier approximation

For $f(x) = x - \pi$ on $[0, 2\pi]$,

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} (x - \pi) dx = 0,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} (x - \pi) \cos(nx) dx = 0,$$

and integration by parts gives

$$b_n = \frac{1}{\pi} \int_0^{2\pi} (x - \pi) \sin(nx) dx = -\frac{2}{n}.$$

Hence the third-order approximation is

$$S_3(x) = -2 \sin x - \sin 2x - \frac{2}{3} \sin 3x.$$

Chapter terminology

Norm	Length induced by an inner product.
Unit vector	Vector of norm one.
Dot product	Euclidean inner product on $\mathbb{R}^n$.
Orthogonal	Inner product equals zero.
Inner product space	Vector space equipped with an inner product.
Orthogonal projection	Closest vector in a specified subspace.
Orthonormal basis	Basis of mutually orthogonal unit vectors.
Gram–Schmidt process	Converts a basis into an orthonormal basis.

Orthogonal complement	complete	Vectors orthogonal to an entire subspace.
Normal equations		$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$
Residual		$\mathbf{b} - A \hat{\mathbf{x}}.$
Cross product		Perpendicular vector in $\mathbb{R}^3$.
Continuous least squares		Function-space projection.
Fourier approximation		Trigonometric least-squares approximation.